# Advanced Git Topics – Cherry-Picking, Tagging and Best Practices

This document explains important Git topics used in professional software development workflows:

1. Cherry-picking commits
2. Tagging versions in Git
3. Writing meaningful commit messages
4. Structuring repositories effectively
5. Using branches for feature development

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## 1. Cherry-Picking Commits

### Definition

Cherry-picking means selecting a specific commit from one branch and applying only that commit to another branch.

It copies one chosen change without merging the complete branch.

### Why Cherry-Picking is Used

- Urgent bug fix needs to move to main branch quickly
- One useful feature commit is needed from another branch
- Avoid merging unfinished work from a branch

### Commands

View commits:

```
git log --oneline
```

Example output:

```
ab12cd3 Fixed login validation bug
ef45gh6 Added dashboard design
```

Apply selected commit to current branch:

```
git cherry-pick ab12cd3
```

### Real Example

A bug was fixed in `feature-login` branch. Only that bug fix is needed in `main`.

```
git checkout main
git cherry-pick ab12cd3
```

Now the bug fix is added to main without merging the full branch.

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## 2. Tagging Versions in Git

### Definition

A tag is a permanent label attached to an important commit, usually used for software releases.

Examples:

- v1.0
- v2.0
- beta-release
- stable-release

### Why Tags are Important

- Mark final release versions
- Easy navigation to old stable versions
- Better release management
- Useful in deployment pipelines

### Commands

Create lightweight tag:

```
git tag v1.0
```

Create annotated tag:

```
git tag -a v1.1 -m "Version 1.1 stable release"
```

View all tags:

```
git tag
```

Push one tag:

```
git push origin v1.0
```

Push all tags:

```
git push origin --tags
```

### Real Example

```
git commit -m "Completed payment module"
git tag -a v2.0 -m "Payment module release"
git push origin main --tags
```

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## 3. Writing Meaningful Commit Messages

### Definition

A commit message should clearly explain what was changed.

### Good Rules

- Use action verbs
- Be specific
- Keep concise but meaningful
- Mention feature or fix

### Good Examples

```
git commit -m "Added student login authentication"
git commit -m "Fixed image upload validation"
git commit -m "Created responsive navbar"
```

### Poor Examples

```
git commit -m "update"
git commit -m "done"
git commit -m "changes"
```

### Better Format

```
Added user dashboard charts
Fixed footer alignment issue
Updated API error handling
```

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## 4. Structuring Repositories Effectively

### Definition

A repository should be organized so any developer can understand files quickly.

### Recommended Structure

```
project-name/
├── src/
```

```
├── public/
├── docs/
├── tests/
├── assets/
├── README.md
└── .gitignore
```

### Why Good Structure Matters

- Faster development
- Easier maintenance
- Better teamwork
- Professional project appearance

### Commands Example

```
mkdir src public docs tests assets
touch README.md .gitignore
```

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## 5. Using Branches for Feature Development

### Definition

A branch is a separate workspace used to build features safely.

### Why Branches are Important

- Main branch remains stable
- Multiple developers can work together
- Features can be tested separately
- Pull Requests become clean

### Commands

Create and switch branch:

```
git checkout -b feature-login
```

Make changes and commit:

```
git add .
git commit -m "Implemented login system"
```

Push branch:

```
git push origin feature-login
```

Merge later:

```
git checkout main
git merge feature-login
```

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## 6. Real Development Scenario

Suppose a team is building an E-commerce project.

- Main branch contains stable website
- Developer A creates `feature-cart`
- Developer B creates `feature-payment`
- Bug fix commit from payment is urgently needed in main
- Cherry-pick is used
- After testing, version is tagged as v1.0

This is how real teams use these concepts.

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## 7. Complete Practical Command Flow

```
mkdir ecommerce-project                 # create project folder
cd ecommerce-project                    # move inside folder
git init                                # initialize repository

mkdir src public docs tests assets      # create folders
touch README.md .gitignore index.html   # create files

git add .                               # stage all files
git commit -m "Initial project setup"  # first commit

git checkout -b feature-login           # create login branch
# make login changes

git add .
git commit -m "Added login feature"
git push origin feature-login

git checkout main                       # switch to main
git checkout -b feature-payment         # payment branch
# make payment changes

git add .
git commit -m "Added payment feature"

# copy only login fix commit if needed
git cherry-pick <commit-hash>
```

```
git checkout main
git merge feature-login                # merge login branch
git merge feature-payment              # merge payment branch

git tag -a v1.0 -m "First stable release"
git push origin main --tags
```

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## 8. Quick Revision Commands

```
git cherry-pick <hash>
git tag v1.0
git tag -a v1.1 -m "release"
git push origin --tags
git checkout -b feature-name
git merge feature-name
git commit -m "Meaningful message"
```

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## Final Understanding

- Cherry-pick copies one selected commit.
- Tags mark versions and releases.
- Good commit messages improve project history.
- Good structure improves readability.
- Branches enable safe parallel development.

These are core practices used in modern Git workflows.